Interpolate Paleo Community Data with Age Uncertainty

Description

Interpolates paleo community proportions to a regular time grid while incorporating per-sample age-model iterations for fossil pollen archives. Datasets without age-uncertainty records are interpolated using consensus ages. Fossil pollen archives are interpolated across their age-model iterations and reduced to the median proportion at each grid point.

Usage

```
interpolate_paleo_community_with_age_uncertainty(

  data_community,

  data_age_uncertainty,

  n_cores = 1L,

  max_expanded_rows = 4e+06,

  ...

)
```

Arguments

`data_community`	A data frame with columns 'dataset_name', 'sample_name', 'taxon', 'age', and 'value'. Must already be in proportion form — see [prepare_community_proportions()].
`data_age_uncertainty`	A tibble produced by [extract_age_uncertainty_from_vegvault()], with columns 'dataset_name' (character), 'sample_name' (character), 'iteration' (integer), and 'age_uncertainty' (double). Rows for dataset names absent from 'data_community' are silently ignored.
`n_cores`	Number of workers to use for consensus-age interpolation. Fossil-core parallelism is handled by dynamic 'targets' branches in the paleo pipeline; use '1' for branched fossil inputs.
`max_expanded_rows`	Maximum approximate number of expanded fossil observation rows processed in one interpolation batch (default: '4e6'). Lower values reduce peak memory use at the cost of more interpolation calls.

...	Additional arguments passed to [interpolate_grouped_time_series()], such as 'time_step', 'age_min', and 'age_max'.

Details

Datasets present in 'data_community' that have no matching rows in 'data_age_uncertainty' are treated as gridpoints and interpolated using consensus ages. This ensures the function degrades gracefully when age uncertainty data are partially available.

The uncertainty-aware interpolation proceeds as follows for each fossil core: 1. The consensus 'age' column is dropped and replaced with 'age_uncertainty' (renamed to 'age') from 'data_age_uncertainty', expanding the data to one row per original observation per age-model iteration. 2. [interpolate_grouped_time_series()] is called grouped by 'c("dataset_name", "taxon", "iteration")'. 3. The median 'value' across iterations is computed for each '(dataset_name, taxon, age)' combination.

Fossil cores are joined and interpolated one dataset at a time, in bounded batches of age-model iterations, before each core is reduced. This bounds the expansion in continental runs, where joining all cores simultaneously can exceed available memory.

In the paleo pipelines, inputs are split into one dynamic target branch per dataset before this function is called. 'targets' therefore caches reduced core outputs and schedules parallel work externally.

Value

A data frame with columns 'dataset_name', 'taxon', 'age', and 'value' at regular time intervals. For fossil pollen archive datasets, 'value' is the median across age-model iterations.

See Also

[extract_age_uncertainty_from_vegvault()], [interpolate_grouped_time_series()]

Examples

```
data_community <- tibble::tibble(

  dataset_name = base::rep("core_a", 2L),

  sample_name = base::c("sample_1", "sample_2"),

  taxon = base::rep("Taxon", 2L),

  age = base::c(0, 100),

  value = base::c(0.25, 0.75)

)

data_age_uncertainty <- tibble::tibble(

  dataset_name = base::character(),

  sample_name = base::character(),
```

```
  iteration = base::integer(),
  age_uncertainty = base::numeric()
)
interpolate_paleo_community_with_age_uncertainty(
  data_community = data_community,
  data_age_uncertainty = data_age_uncertainty,
  age_min = 0,
  age_max = 100,
  time_step = 100
)
```